Artificial Intelligence
Lab 1
Introduction to Prolog (https://swish.swi-prolog.org)

Prolog (short for **"PROgramming in LOGic"**) is a **logic programming language** used for:

- Artificial Intelligence (AI)
- Expert systems
- Natural language processing
- Knowledge representation and reasoning

It was created in the **1970s** and is very different from languages like Python or C — it focuses on **what is true**, not how to compute it.

Concept	Description
Facts	Known truths. e.g., parent(john, mary).
Rules	Logical relationships. e.g., grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
Queries	Questions you ask. e.g., ?- parent(john, X).
Backtracking	Automatically tries multiple possibilities to find all answers

Facts:
In Prolog, a fact is a basic assertion(declaration) about some world knowledge. It declares that something is unconditionally true.

Syntax of a Fact:
predicate(argument1, argument2, ..., argumentN).

- `predicate` describes the relationship.
- `arguments` are objects or entities.
- Facts must end with a **period (.)**

Example Facts

parent(john, mary).
male(john).
female(mary).
likes(alice, pizza).

This mean:

John is a parent of Mary
John is male

Mary is female
Alice likes pizza

How Prolog Uses Facts?

You can ask questions (queries) like:

?- parent(john, mary).

Prolog will answer:

true.

?- parent(john, X).

Prolog answers:

X = mary.

Lab : Family Relationship

```
parent(john, mary).
parent(john, paul).
sibling(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.
```

In the query box (bottom right), type:

```
?- sibling(mary, X).
```

Click “Run!”

Result:

```
X = paul
```

Write a Prolog program that defines a simple family tree and lets you query multiple family relationships like:

parent
father / mother
sibling
grandparent
uncle / aunt
cousin

Prolog Program

```
% ---------- Basic Facts ----------
male(john).
male(paul).
male(tom).
male(alex).

female(mary).
female(susan).
female(linda).
female(jane).

parent(john, paul).
parent(john, mary).
parent(mary, tom).
parent(mary, susan).
parent(paul, alex).
parent(paul, jane).

% ---------- Basic Relationships ----------
father(X, Y) :- parent(X, Y), male(X).
mother(X, Y) :- parent(X, Y), female(X).

% ---------- Siblings ----------
sibling(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.

brother(X, Y) :- sibling(X, Y), male(X).
sister(X, Y) :- sibling(X, Y), female(X).

% ---------- Grandparents ----------
grandparent(X, Y) :- parent(X, Z), parent(Z, Y).
grandfather(X, Y) :- grandparent(X, Y), male(X).
grandmother(X, Y) :- grandparent(X, Y), female(X).

% ---------- Uncle and Aunt ----------
uncle(X, Y) :- brother(X, Z), parent(Z, Y).
aunt(X, Y) :- sister(X, Z), parent(Z, Y).

% ---------- Cousins ----------
cousin(X, Y) :-
    parent(P1, X),
    parent(P2, Y),
    sibling(P1, P2),
    X \= Y.
```

Example Queries You Can Run

?- father(john, paul).
?- sibling(mary, paul).
?- grandfather(john, tom).
?- uncle(paul, tom).
?- cousin(tom, alex).

Example Queries You Can Run

?- cousin(tom, X).

Assignment 1

Title: Family Relationship in Prolog

Objective:

To model a family tree using facts and rules in Prolog, and write queries to infer relationships like:

Parent, Sibling, Grandparent
Uncle, Aunt, Cousin
Father, Mother
Brother, Sister

Instructions

Create a Prolog program named family.pl.

Define:

At least 8 individuals
Their genders
Parent-child relationships

Write rules for:

father, mother
sibling, brother, sister
grandparent, grandfather, grandmother
uncle, aunt
cousin

Run and test the following queries:

Who are the siblings of Mary?
Who is John's granddaughter?
Does Paul have a cousin?
Who are the uncles of Tom?